

Phase 2 — AR/ARMA/ARFIMA: Time Series Benchmark

Complete Analysis of Results

Quantitative Energy Research
Mathematical Finance RAship

Input: `W_tilde_series` from Phase 1
Sample: 1,462 daily observations, 2015–2018

Šaltytė Benth & Benth (2008) · Alexandrinis (2013) · Torres et al. (2005) · Ailliot, Monbet & Prevosto (2006)

Contents

1 Overview and Purpose of Phase 2	2
2 Data Import (Cell 1)	2
3 ACF/PACF and AR Order Identification (Cell 2)	2
3.1 Theoretical background	2
3.2 ACF and PACF table	3
3.3 Interpretation of the PACF	3
3.4 AIC grid search	4
4 AR(4) Estimation (Cell 3)	4
4.1 Model specification	4
4.2 Estimated coefficients	5
4.3 Interpretation	5
5 Residual Diagnostics (Cell 4)	6
5.1 Ljung-Box on $\hat{\varepsilon}(t)$	6
5.2 Ljung-Box on $\hat{\varepsilon}^2(t)$ and ARCH-LM test	6
5.3 Residual normality	7
6 ARMA Grid Search (Cell 5)	7
6.1 Motivation	7
6.2 Full grid results	7
6.3 Interpretation	8
7 Long Memory Tests (Cell 7)	8
7.1 Motivation and tests	8
7.2 GPH log-periodogram estimator	8
7.3 R/S Hurst exponent	8
7.4 Decision	9
8 Time-Varying AR Coefficients (Cell 9)	9
8.1 Motivation	9
8.2 Results	9
8.3 Interpretation	9
9 Out-of-Sample Evaluation (Cell 10)	10
9.1 Walk-forward design	10
9.2 Results	10
10 Phase 2 Complete Summary	10

Overview and Purpose of Phase 2

Phase 2 operates exclusively on the standardised series $\tilde{W}(t)$ produced by Phase 1. The full preprocessing chain of Phase 1—Box-Cox power transformation ($\lambda = 0.1$), removal of the seasonal mean $\Lambda(t)$ via a two-harmonic Fourier series, and removal of the seasonal variance $\sigma^2(t)$ via a cosine Fourier series—delivered a series that is zero-mean ($\tilde{W} = 0.001$), unit-variance ($s = 0.967$), and free of all deterministic seasonal structure. What remains is the stochastic weather signal: the short-term atmospheric memory that makes today's wind speed anomaly informative about tomorrow's.

Phase 2 characterises and models this memory through five logical steps:

1. Identify the AR order from the PACF and an AIC grid search over $\text{AR}(p)$, $p = 1, \dots, 6$.
2. Estimate $\text{AR}(4)$ by OLS and validate residuals with the Ljung-Box test (serial autocorrelation) and the ARCH-LM test (conditional heteroskedasticity).
3. Compare $\text{AR}(4)$ against a full $\text{ARMA}(p, q)$ grid ($p \leq 5$, $q \leq 2$) to confirm that no MA terms are needed.
4. Test for long memory via the GPH log-periodogram estimator and the R/S Hurst exponent; decide whether ARFIMA is required.
5. Assess whether the AR coefficients vary seasonally using rolling estimation and Fourier F-tests.

The phase also includes a walk-forward out-of-sample evaluation and a Diebold-Mariano test to quantify the forecasting gain over the naive persistence benchmark. The key output is the fitted $\text{AR}(4)$ model object `fit_ar4` and the coefficient file `ar4.coefficients.csv`, which Phase 4 will use to construct the $\text{CAR}(4)$ companion matrix.

Data Import (Cell 1)

The notebook loads $\tilde{W}(t)$ from `W_tilde_phase1.csv`, the CSV file written at the end of Phase 1:

```
W_tilde_series = pd.read_csv("W_tilde_phase1.csv",
                             index_col=0, parse_dates=True)
W_tilde_series = W_tilde_series.squeeze()
```

The `squeeze()` call converts the single-column DataFrame back to a pandas Series, as required by all downstream `statsmodels` routines. Cell 1 confirms 1,462 observations with date range 2014-12-31 to 2018-12-31, exactly consistent with Phase 1 output.

ACF/PACF and AR Order Identification (Cell 2)

Theoretical background

For a pure $\text{AR}(p)$ process, the partial autocorrelation function (PACF) is exactly zero at all lags $k > p$ and non-zero at lags $1, \dots, p$. This hard cutoff makes the PACF the standard visual diagnostic for AR order selection. The autocorrelation function (ACF), by contrast, decays geometrically for any AR process and never cuts off sharply; it is used here to confirm the absence of remaining seasonal structure rather than to identify the

model order. The 95% significance bound for both statistics under the white-noise null hypothesis is:

$$\pm \frac{1.96}{\sqrt{n}} = \pm \frac{1.96}{\sqrt{1,462}} = \pm 0.0513.$$

ACF and PACF table

Table 1: ACF and PACF of $\tilde{W}(t)$ at selected lags. * * * denotes significance at the 5% level ($|\text{value}| > 0.0513$). Lags 9–17 are omitted as all are non-significant in both columns.

Lag	ACF Sig.	PACF Sig.
1	+0.5792 ***	+0.5796 ***
2	+0.2830 ***	−0.0791 ***
3	+0.2053 ***	+0.1126 ***
4	+0.1731 ***	+0.0311
5	+0.1424 ***	+0.0290
6	+0.1035 ***	−0.0013
7	+0.0821 ***	+0.0169
8	+0.0509	−0.0218
<i>(lags 9–17: all non-significant)</i>		
18	+0.0563 ***	+0.0746 ***
19	+0.0822 ***	+0.0277
20	+0.0894 ***	+0.0306

Interpretation of the PACF

Lags 1, 2, and 3 are clearly significant in the PACF (+0.580, −0.079, +0.113). Lag 4 (= 0.031) falls below the 0.0513 threshold by a small margin, and lags 5 onwards are comfortably non-significant. This gives a borderline AR(3)/AR(4) visual reading. This is consistent with Alexandrinis (2013), who reports an almost identical pattern for New York DAWS where $\hat{\beta}_4$ has a p -value of 0.066 in his 20-year sample—also borderline. The AIC grid resolves the ambiguity formally.

The mild return to significance at lags 18–20 does not indicate a higher-order AR process. With 20 simultaneous tests at the 5% level, the probability of at least one spurious rejection is $1 - 0.95^{20} \approx 64\%$. Additionally, lags 18–20 days correspond to the characteristic period of Rossby planetary wave trains and synoptic blocking events in the mid-latitude atmosphere, a genuine physical periodicity that does not call for higher AR lags.

AIC grid search

Table 2: AIC grid search: $\text{AR}(p)$, $p = 1, \dots, 6$. Lower AIC denotes better fit penalised for model complexity. The arrow ($\leftarrow$) marks the AIC minimum.

Model	Log-likelihood	AIC	BIC
AR(1)	-1725.96	3455.919	3466.494
AR(2)	-1721.40	3448.806	3464.669
AR(3)	-1712.12	3432.244	3453.394 $\leftarrow$
AR(4)	-1711.42	3432.832	3459.270
AR(5)	-1710.80	3433.591	3465.317
AR(6)	-1710.79	3435.590	3472.603

The AIC minimum is at AR(3) (AIC = 3432.244), with AR(4) only 0.59 units higher. By the Burnham & Anderson criterion, models within 2 AIC units of the minimum are statistically indistinguishable. AR(3) and AR(4) are therefore practically equivalent on this dataset.

AR(4) is retained as the canonical model. The justification is structural rather than statistical. The continuous-time CAR(4) framework established by Šaltytė Benth & Benth (2008) and Benth & Šaltytė Benth (2009) requires exactly four AR lags because the companion matrix A is 4×4 and the Euler discretisation mapping $\{\beta_1, \dots, \beta_4\} \mapsto \{\alpha_1, \dots, \alpha_4\}$ uses all four coefficients. Adopting AR(3) would produce a CAR(3) specification with different stationarity properties and pricing formulas. The 0.59-unit AIC advantage of AR(3) does not justify this theoretical deviation from the established literature framework.

AR(4) Estimation (Cell 3)

Model specification

The AR(4) model for $\tilde{W}(t)$ is estimated without an intercept because $\tilde{W}(t)$ is zero-mean by construction:

$$\tilde{W}(t) = \beta_1 \tilde{W}(t-1) + \beta_2 \tilde{W}(t-2) + \beta_3 \tilde{W}(t-3) + \beta_4 \tilde{W}(t-4) + \varepsilon(t). \quad (1)$$

Parameters are estimated by OLS, equivalent to conditional maximum likelihood under Gaussian innovations. The `ar_idx` filter ensures only the four AR rows are extracted from the `statsmodels` parameter vector, excluding the appended `sigma2` variance term.

Estimated coefficients

Table 3: AR(4) OLS estimates. Significance: *** $p < 0.01$, ** $p < 0.05$. Literature benchmark from Alexandrinis (2013, Table 3), New York DAWS.

Param.	Estimate	Std. Err.	t -stat	p -value	Sig.
$\hat{\beta}_1$	+0.6299	0.0233	+26.98	< 0.001	***
$\hat{\beta}_2$	−0.1443	0.0292	−4.945	< 0.001	***
$\hat{\beta}_3$	+0.0926	0.0280	+3.301	0.001	***
$\hat{\beta}_4$	+0.0310	0.0246	+1.260	0.208	
Log-likelihood = −1711.42; AIC = 3432.83; BIC = 3459.27					
<i>Literature (Alexandrinis 2013):</i> $\beta_1 = +0.362$, $\beta_2 = -0.100$, $\beta_3 = +0.027$, $\beta_4 = +0.022$					

Interpretation

Sign pattern. The coefficient vector has signs (+, −, +, +), identical to the New York results of both Šaltytė Benth & Benth (2008) and Alexandrinis (2013). The positive $\hat{\beta}_1 = +0.630$ captures persistence: 63% of yesterday’s wind anomaly carries forward to today. The negative $\hat{\beta}_2 = -0.144$ captures damped oscillation: above-average wind is followed by below-average wind then a partial recovery, the signature of synoptic weather systems passing through the measurement location. The smaller positive $\hat{\beta}_3$ and $\hat{\beta}_4$ represent the multi-day relaxation tail.

Magnitude vs. literature. Coefficients are larger than New York because the ACF at lag 1 (= 0.579) is substantially higher than for New York (≈ 0.36). The proxy dataset at approximately 10 m height in a flat energy-market context exhibits stronger synoptic-scale persistence than the coastal New York site, where orographic variability introduces additional sub-synoptic noise.

Stationarity. The four characteristic roots of (1) have moduli 0.702, 0.470, 0.470, and 0.200, all strictly less than one. The process is covariance-stationary, a necessary condition for the Gaussian CAR(4) embedding in Phase 4.

Mean-reversion half-life. The dominant root (modulus 0.702) determines the rate at which the impulse response decays. The implied half-life is:

$$\tau_{1/2} = \frac{-\ln 2}{\ln(0.702)} \approx 2.0 \text{ days.} \quad (2)$$

This is physically consistent with the typical residence time of mid-latitude synoptic weather systems at a fixed observation point.

Residual Diagnostics (Cell 4)

Ljung-Box on $\hat{\varepsilon}(t)$

The Ljung-Box Q-statistic tests the joint null hypothesis that all autocorrelations of the residuals up to lag m are zero:

$$\text{LBQ}(m) = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}^2(k)}{n-k} \sim \chi^2(m) \quad \text{under } H_0. \quad (3)$$

Table 4: Ljung-Box test on AR(4) residuals $\hat{\varepsilon}(t)$. H_0 : no serial autocorrelation. Pass requires $p > 0.05$.

Lags	LBQ stat.	p -value	Verdict
5	1.196	0.945	Pass
10	2.884	0.984	Pass
15	9.334	0.860	Pass
20	19.035	0.520	Pass

The AR(4) has completely removed all serial autocorrelation from $\tilde{W}(t)$. The p -value of 0.945 at lag 5 represents an overwhelming failure to reject the white-noise null, consistent with Alexandrinis (2013) who reports LBQ $p = 0.989$ at all lags after fitting AR(4) to New York DAWS.

Ljung-Box on $\hat{\varepsilon}^2(t)$ and ARCH-LM test

Table 5: Ljung-Box test on squared AR(4) residuals $\hat{\varepsilon}^2(t)$. H_0 : homoskedastic residuals. $p < 0.05$ indicates ARCH effects.

Lags	LBQ stat.	p -value	Verdict
5	22.961	0.0003	ARCH effects
10	31.789	0.0004	ARCH effects
15	36.663	0.0014	ARCH effects
20	41.313	0.0034	ARCH effects

The Engle (1982) ARCH-LM test reinforces this conclusion:

$$\begin{aligned} \text{ARCH-LM}(5) &= 18.80, & p &= 0.0021, & R^2 &= 0.013, \\ \text{ARCH-LM}(10) &= 23.23, & p &= 0.0099, & R^2 &= 0.016. \end{aligned}$$

The AR(4) residuals are serially uncorrelated *in mean*, but their *variance* clusters over time: high-magnitude residuals tend to follow other high-magnitude residuals. This stochastic volatility clustering is distinct from the deterministic seasonal variance $\sigma^2(t)$ removed in Phase 1. Tol (1997) was the first to identify ARCH effects in daily wind speed residuals and to argue for GARCH models as the appropriate extension. These results constitute the direct empirical motivation for Phase 3.

Residual normality

The AR(4) residuals have skewness = +0.361, Pearson kurtosis = 4.129, JB stat = 109.39 ($p = 0.000$), and KS $p = 0.083$. The KS test passes at the 5% level, suggesting approximate normality in the body of the distribution, while the JB test detects the tail excess. This departure from normality is smaller than in $\tilde{W}(t)$ (skewness +0.511, kurtosis 3.925) and is expected: the residual non-normality is largely generated by the GARCH clustering, which inflates the unconditional kurtosis even when the conditional distribution is Gaussian. Once GARCH is fitted in Phase 3 and residuals are divided by the estimated conditional standard deviation $\hat{h}(t)^{1/2}$, the standardised GARCH residuals are expected to be approximately normal.

Diagnostic summary. The AR(4) residuals are (a) serially uncorrelated in mean, (b) conditionally heteroskedastic with confirmed ARCH effects, and (c) mildly non-normal due to volatility clustering. This is precisely the residual structure described by Tol (1997) and Alexandrinis (2013) as the entry point for the GARCH extension. Phase 2 is complete and Phase 3 is correctly motivated.

ARMA Grid Search (Cell 5)

Motivation

Although the PACF and AIC grid both point to a pure AR specification, moving average terms could in principle absorb short-lived shocks that the AR structure handles sub-optimally. Following Torres et al. (2005), all ARMA(p, q) models with $p \in \{1, \dots, 5\}$ and $q \in \{0, 1, 2\}$ are estimated and compared by AIC.

Full grid results

Table 6: ARMA(p, q) grid search, $p \leq 5$, $q \leq 2$. ARMA(1,2) achieves the global AIC minimum; AR(4) is the literature benchmark.

Model	Log-lik.	AIC	BIC	Note
AR(1)	−1725.96	3455.92	3466.49	
ARMA(1,1)	−1719.13	3444.26	3460.13	
ARMA(1,2)	−1711.55	3431.11	3452.26	AIC best
AR(2)	−1721.40	3448.81	3464.67	
ARMA(2,1)	−1716.21	3440.43	3461.58	
ARMA(2,2)	−1711.17	3432.33	3458.77	
AR(3)	−1712.12	3432.24	3453.39	
ARMA(3,1)	−1711.02	3432.03	3458.47	
ARMA(3,2)	−1710.91	3433.81	3465.54	
AR(4)	−1711.42	3432.83	3459.27	Literature bench.
ARMA(4,1)	−1710.95	3433.89	3465.62	
ARMA(4,2)	−1710.27	3434.54	3471.55	
AR(5)	−1710.80	3433.59	3465.32	
ARMA(5,1)	−1710.40	3434.81	3471.82	
ARMA(5,2)	−1710.24	3436.49	3478.79	

Interpretation

The global AIC minimum is ARMA(1,2) (AIC = 3431.11), with:

$$\Delta\text{AIC} = 3431.11 - 3432.83 = -1.73.$$

Since $|\Delta\text{AIC}| < 2$, ARMA(1,2) and AR(4) are statistically indistinguishable. This outcome confirms the finding of Torres et al. (2005): MA terms provide no statistically meaningful improvement over a well-specified pure AR model for daily wind speed data.

AR(4) is retained as the canonical model on two grounds. First, the practical equivalence implies no meaningful gain in fit or forecasting accuracy. Second, the CAR(4) theoretical framework requires a pure AR(4) discrete-time representation; the Euler discretisation mapping $\{\beta_1, \dots, \beta_4\} \mapsto \{\alpha_1, \dots, \alpha_4\}$ used in Phase 4 does not generalise to ARMA models.

Long Memory Tests (Cell 7)

Motivation and tests

Standard AR/ARMA models assume *short memory*: past shocks decay geometrically. If wind speed exhibits long memory—a hyperbolic autocorrelation decay persisting over very long horizons—the appropriate model is ARFIMA(p, d, q) with fractional differencing parameter $d \in (0, 0.5)$. Two independent diagnostics are applied.

GPH log-periodogram estimator

The Geweke & Porter-Hudak (1983) estimator regresses the log periodogram $\ln I(\omega_j)$ on $\ln |2 \sin(\omega_j \pi)|$ at the $m = n^{0.6} = 79$ lowest Fourier frequencies. The slope equals $-2\hat{d}$, so $\hat{d} = -\frac{1}{2}\hat{\beta}_{\text{slope}}$. Results:

$$\begin{aligned}\hat{d} &= 0.1188, \\ \text{SE} &= 0.0936, \\ t\text{-stat} &= 1.270, \\ p\text{-value} &= 0.208.\end{aligned}$$

The null $H_0 : d = 0$ cannot be rejected. The 95% confidence interval $[-0.065, 0.303]$ comfortably includes zero.

R/S Hurst exponent

The classical rescaled-range statistic yields $\hat{H} = 0.727$ (SE = 0.013), implying $\hat{d} = \hat{H} - 0.5 = 0.227$. This appears to contradict the GPH result, but the contradiction is spurious. The R/S estimator is sensitive to strong short-range autocorrelation: a dominant AR(1) coefficient of $\hat{\beta}_1 = 0.630$ inflates the estimated Hurst exponent in finite samples even for a process with no genuine long memory, because the cumulative deviation range grows faster than $\sqrt{n}$ over short sub-windows. The GPH estimator is explicitly designed to isolate the low-frequency spectral slope and is the correct formal test for fractional integration. The R/S result is reported as a complementary diagnostic.

Decision

The GPH test fails to reject $d = 0$ at $p = 0.208$. ARFIMA is not required; AR(4) is the appropriate specification for daily $\tilde{W}(t)$. This is fully consistent with Šaltytė Benth & Benth (2008) and Alexandrinis (2013), who both find AR(4) sufficient for daily DAWS. Long memory in wind speed is primarily a feature of hourly data (cf. Kavasseri & Seetharaman 2009; Haslett & Raftery 1989) and is attenuated by the daily averaging operation.

Time-Varying AR Coefficients (Cell 9)

Motivation

Ailliot, Monbet & Prevosto (2006) propose AR models with time-varying coefficients driven by seasonal regime changes in atmospheric circulation patterns. To test whether the AR(4) coefficients in this dataset vary with the season, the model is re-estimated in 157 rolling windows of 365 days each (step 7 days). For each coefficient $\hat{\beta}_k(t)$, an F-test is performed against the null of seasonal constancy, using a Fourier regression of the rolling estimates on the annual and semi-annual harmonics.

Results

Table 7: Seasonal variation F-test for rolling AR(4) coefficients. Window = 365 days, step = 7 days, $n_{\text{windows}} = 157$. Significance: *** $p < 0.01$.

Coeff.	Mean	Range	F -stat	p -value	Sig.
β_1	+0.630	0.173	0.84	0.502	
β_2	−0.131	0.205	4.61	0.002	***
β_3	+0.076	0.301	3.59	0.008	***
β_4	+0.049	0.205	7.10	< 0.001	***

Interpretation

The results reveal a clear and physically meaningful asymmetry. β_1 —the dominant lag-1 persistence coefficient—is *constant across seasons* ($F = 0.84$, $p = 0.502$), reflecting the fact that the fundamental atmospheric memory timescale is a year-round property of the large-scale circulation.

By contrast, β_2 , β_3 , and β_4 —the correction and oscillatory terms—all vary significantly with the season. Strikingly, β_4 is the *most* seasonally variable ($F = 7.10$, $p < 0.001$), despite being globally insignificant in the full-sample fixed-coefficient estimate ($p = 0.208$ in Table 3). This apparent contradiction has a precise explanation: β_4 is genuinely non-zero in certain seasons but near-zero in others, so its full-sample average is small and imprecisely estimated. This is exactly the mechanism Ailliot et al. (2006) motivate: a coefficient that appears globally redundant but carries real explanatory power in specific atmospheric regimes.

This finding has a direct implication for Phase 4. The seasonal variation of β_2 , β_3 , β_4 suggests that the CAR(4) mean-reversion coefficients α_k may benefit from being modelled as Fourier series of t , paralleling the time-varying framework of Ailliot et al. (2006).

Out-of-Sample Evaluation (Cell 10)

Walk-forward design

The dataset is split 85/15 into a training set (1,242 observations) and a test set (220 observations). A walk-forward scheme refits each model sequentially on all observations up to time t , generates $\hat{W}(t+1|t)$, and advances. Three models are compared: persistence (naive), AR(4), and ARMA(1,2). The Diebold-Mariano (1995) test formally tests equal predictive accuracy between AR(4) and persistence.

Results

Table 8: One-step-ahead out-of-sample performance. Test set: 220 observations.

Model	RMSE	MAE
Persistence	0.6856	0.5214
AR(4)	0.6017	0.4706
ARMA(1,2)	0.6022	0.4702

AR(4) achieves a **12.2% RMSE reduction** relative to the persistence benchmark. The Diebold-Mariano test:

$$\begin{aligned} \text{DM stat} &= -3.681, \\ p\text{-value} &= 0.0002. \end{aligned}$$

H_0 (equal predictive accuracy) is rejected at all conventional levels. AR(4) is statistically and practically more accurate than persistence.

The out-of-sample gap between AR(4) and ARMA(1,2) is negligible (RMSE difference = 0.0005, MAE difference < 0.001), providing the definitive out-of-sample confirmation of the AIC grid result: MA terms add no practical forecasting value.

Literature comparison. Torres et al. (2005) report ARMA-based RMSE improvements of 15–25% over persistence for hourly Navarre wind data at 1-hour horizons, decreasing at longer horizons. The 12.2% gain at the 1-day horizon is consistent with this range: daily aggregation smooths the high-frequency variability that AR models capture most effectively at hourly scales, while preserving the synoptic-scale memory that drives the improvement over persistence.

Phase 2 Complete Summary

Complete Phase 2 results.

[1] **AR order identification.** PACF cutoff at lags 1–3 with lag 4 borderline (= 0.031, threshold 0.051). AIC grid minimum at AR(3), with AR(4) only $\Delta\text{AIC} = 0.59$ higher (indistinguishable). AR(4) retained on CAR(4) theoretical grounds.

[2] **AR(4) estimated coefficients.** $\hat{\beta}_1 = +0.6299$ (***), $\hat{\beta}_2 = -0.1443$ (***),

$\hat{\beta}_3 = +0.0926$ (***), $\hat{\beta}_4 = +0.0310$ (ns). Sign pattern $(+, -, +, +)$ consistent with literature. All characteristic root moduli < 1 : stationarity confirmed. Mean-reversion half-life $\tau_{1/2} \approx 2.0$ days. AIC = 3432.83, BIC = 3459.27.

[3] **Residual diagnostics.** LBQ on $\hat{\varepsilon}(t)$: $p(20) = 0.520 \rightarrow$ autocorrelation fully removed $\checkmark$. LBQ on $\hat{\varepsilon}^2(t)$: $p(20) = 0.003 \rightarrow$ ARCH effects present $\rightarrow$ GARCH needed. ARCH-LM(5): $p = 0.002 \rightarrow$ confirms GARCH $\rightarrow$ Phase 3.

[4] **ARMA grid** ($p \leq 5, q \leq 2$). Global best ARMA(1,2), AIC = 3431.11. Δ AIC vs AR(4) = -1.73 ($|\Delta| < 2$: indistinguishable). AR(4) retained for CAR(4) compatibility.

[5] **Long memory.** GPH $\hat{d} = 0.119$ ($p = 0.208$) $\rightarrow$ no significant long memory. Hurst $\hat{H} = 0.727$ inflated by strong AR(1) = 0.630; GPH is the correct formal test. ARFIMA: not required; AR(4) adequate at daily resolution.

[6] **Time-varying coefficients.** β_1 constant across seasons ($F = 0.84, p = 0.502$). $\beta_2, \beta_3, \beta_4$ vary significantly ($p = 0.002, 0.008, < 0.001$). β_4 is the most seasonally variable despite being globally insignificant. Motivates time-varying CAR model of Ailliot et al. (2006) in Phase 4.

[7] **Out-of-sample performance.** AR(4) RMSE = 0.6017 vs. persistence = 0.6856 $\rightarrow$ 12.2% improvement. DM test: $p = 0.0002 \rightarrow$ AR(4) significantly more accurate.

Phase 2 output: `fit_ar4` model object and `ar4_coefficients.csv`. AR(4) coefficients ready for Phase 4 CAR(4) embedding. ARCH effects confirmed $\rightarrow$ Phase 3 (GARCH) is next.